

Properties Of Solids And Liquids

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1

$$\frac{\Delta v}{v} = \frac{\Delta p}{B} = \frac{1}{20}$$
$$\frac{\Delta I}{I} = \frac{1}{3} \frac{\Delta p}{B} = \frac{1}{60}$$
$$\frac{\Delta I}{I} \times 100 = \frac{5}{3} = 1.67\%$$

Q2

$$v_T \propto r^2$$

$$\sqrt{2gh} \propto r^2$$

$$\Rightarrow h \propto r^4$$

Q3

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$k = \frac{YA}{L}$$

$$f = \left(\frac{1}{2\pi} \right) \sqrt{\frac{YA}{mL}}$$

Q4

$$V = 2\alpha_2 + \alpha_1$$
$$= 10 \times 10^{-6} + 5 \times 10^{-5}$$
$$= 60 \times 10^{-6} / ^\circ C$$

Q5

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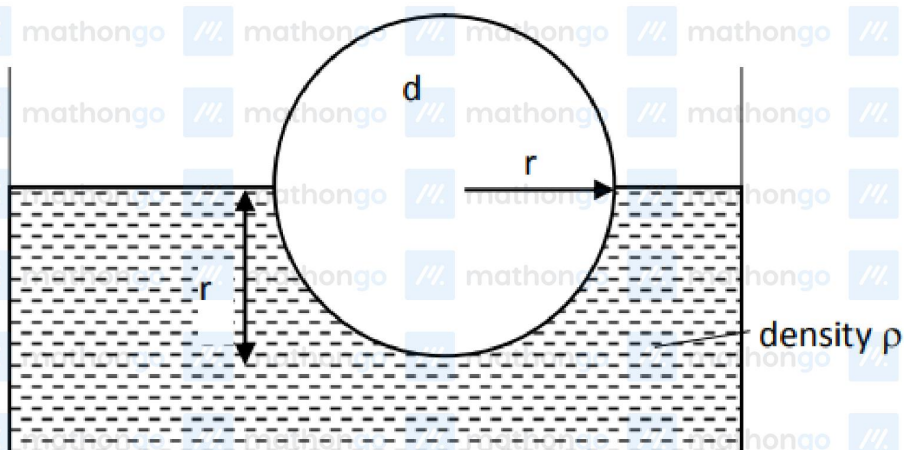
Physics

$$dVg = \rho \left(\frac{V}{2} \right) g + T(2\pi r)$$

$$\Rightarrow d \cdot \frac{4}{3}\pi r^3 g = \rho \cdot \frac{2}{3}\pi r^3 g + 2\pi r T$$

$$\Rightarrow \frac{2}{3}r^2 g (2d - \rho) = 2T$$

$$\Rightarrow r = \sqrt{\frac{3T}{(2d - \rho)g}}$$



Q6

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Physics

$$\frac{du}{dv} = \frac{1}{2} \text{stress} \times \frac{\text{stress}}{y}$$

$$= \frac{1}{2} \frac{F^2}{A^2 y}$$

$$\frac{du}{dv} \propto \frac{1}{d^4}$$

$$\frac{\left(\frac{du}{dv}\right)_1}{\left(\frac{du}{dv}\right)_2} = \frac{d_2^4}{d_1^4} = \frac{1}{4}$$

$$\frac{d_1}{d_2} = (4)^{1/4}$$

$$\frac{d_1}{d_2} = \sqrt{2} : 1$$